

Minimax Estimation

Outline

- 1) Minimax risk, estimator
- 2) Least favorable priors
- 3) Examples

Minimax risk

Last idea for choosing an estimator: worst-case risk

$$\underset{\delta}{\text{minimize}} \quad \sup_{\theta} R(\theta; \delta)$$

The minimum achievable sup-risk is called the minimax risk of the estimation problem

$$r^* = \inf_{\delta} \sup_{\theta} R(\theta; \delta)$$

An estimator δ^* is called minimax if it achieves the minimax risk, i.e.

$$\sup_{\theta} R(\theta; \delta^*) = r^*$$

Game theory interpretation: (Minimax game)

- 1) Analyst chooses estimator δ
- 2) Nature chooses parameter θ to max. risk

NB: Nature chooses θ adversarially, not X

Compare to Bayes ("Maximin" game)

- 1) Nature chooses prior $\triangleleft$ (mixed strategy)
- 2) Analyst chooses estimator to min. (avg) risk

We will look for Nature's Nash-equil. strategy in this problem

Least Favorable Priors

Key observation: average-case risk $\leq$ worst-case risk

For proper prior Λ , the Bayes risk is

$$r_{\Lambda} = \inf_{\delta} \int R(\theta; \delta) d\Lambda(\theta) \\ \leq \inf_{\delta} \sup_{\theta} R(\theta; \delta) = r^*$$

If δ_{Λ} Bayes then $r_{\Lambda} = \int R(\theta; \delta_{\Lambda}) d\Lambda(\theta)$

$\Rightarrow$ Bayes risk of any Bayes estimator
lower bounds r^*

Least favorable prior Λ^* gives best
lower bound: $r_{\Lambda^*} = \sup_{\Lambda} r_{\Lambda}$

Sup-risk of any estimator upper bounds r^*

$$r_{\Lambda} \leq r_{\Lambda^*} \leq r^* \leq \sup_{\theta} R(\theta; \delta) \\ \begin{matrix} \uparrow & & \uparrow \\ \text{(any } \Lambda) & & \text{(any } \delta) \end{matrix}$$

Idea: try to match upper & lower bounds

Theorem If $r_{\Delta} = \sup_{\theta} R(\theta; \delta_{\Delta})$ with
Bayes estimator δ_{Δ} then:

(a) δ_{Δ} is minimax

(b) If δ_{Δ} is unique Bayes (up to a.s.)
for Δ , it is unique minimax

(c) Δ is least fav.

Proof a) Any other δ :

$$\begin{aligned} \sup_{\theta} R(\theta; \delta) &\geq \int R(\theta; \delta) d\Delta(\theta) \\ &\geq \int R(\theta; \delta_{\Delta}) d\Delta(\theta) \quad (*) \\ &= r_{\Delta} \end{aligned}$$

$$= \sup_{\theta} R(\theta; \delta_{\Delta}) \quad \text{by assumption}$$

$\Rightarrow r_{\Delta}$ is minimax risk, δ_{Δ} is minimax,

b) Replace " $\geq$ " with ">" in 2nd ineq. (*)

c) Any other prior $\tilde{\Delta}$:

$$r_{\tilde{\Delta}} = \inf_{\delta} \int R(\theta; \delta) d\tilde{\Delta}(\theta)$$

$$\leq \int R(\theta; \delta_{\Delta}) d\tilde{\Delta}(\theta)$$

$$\leq \sup_{\theta} R(\theta; \delta_{\Delta}) = r_{\Delta} \quad \square$$

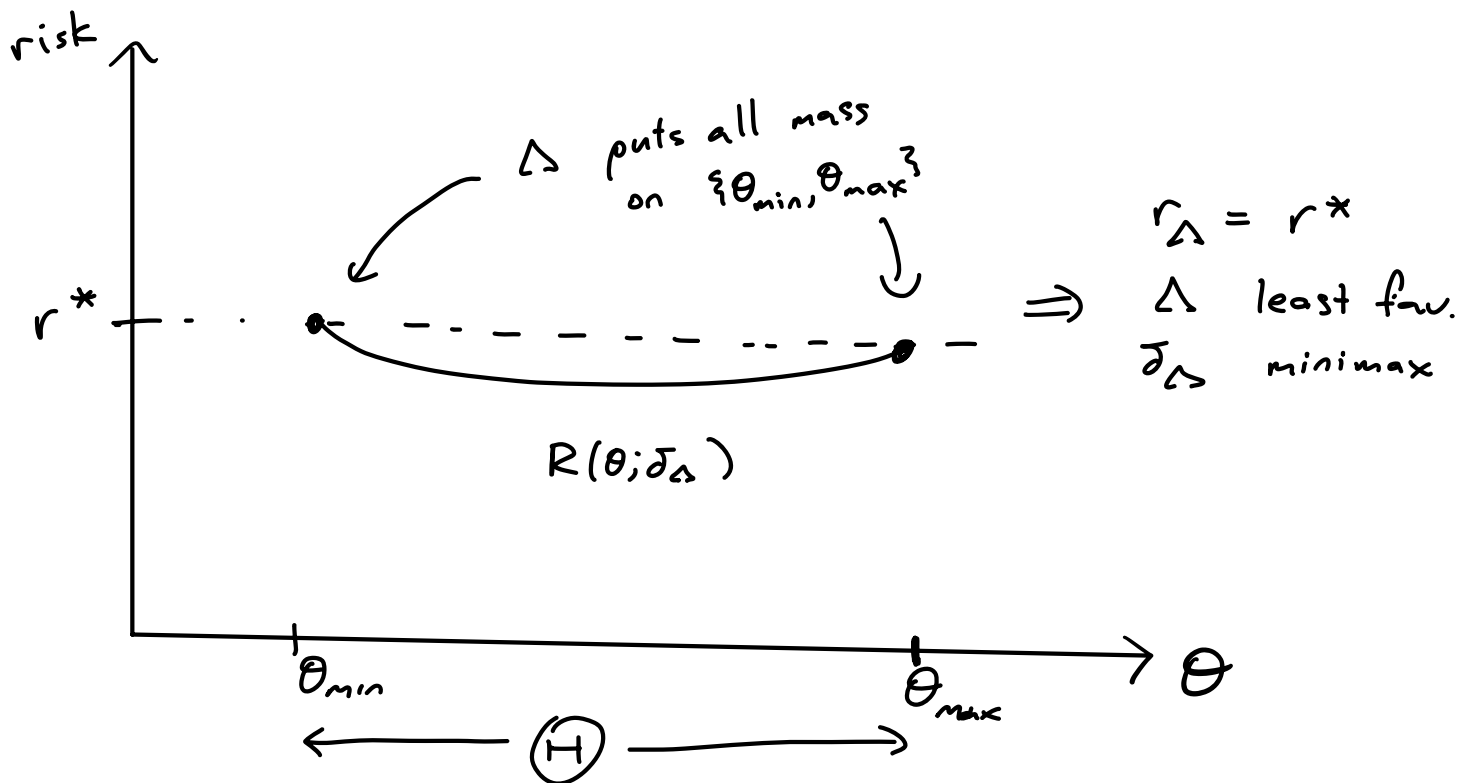
The above theorem gives a checkable condition:
 does $\text{avg risk} = \text{sup risk}$?

True if :

mistake on final: saying r_Δ is const. doesn't prove anything

1) $R(\theta; \delta_\Delta)$ is constant

2) $\Delta(\{\theta : R(\theta; \delta_\Delta) = \max_{\xi} R(\xi; \delta_\Delta)\}) = 1$



Example (Binomial)

$X \sim \text{Binom}(n, \theta)$, estimate θ , sq. err.

Try $\text{Beta}(\alpha, \beta)$, hope to get one with const. risk

$$\delta_{\alpha, \beta}(X) = \frac{\alpha + X}{\alpha + \beta + n} \quad \text{Try } \alpha = \beta \text{ (symmetric)}$$

$$\begin{aligned} \text{MSE}(\theta; \delta_{\alpha, \alpha}) &= \left(\frac{\alpha + \theta n}{2\alpha + n} - \theta \right)^2 + (2\alpha + n)^{-2} \text{Var}_{\theta}(X) \\ &= (2\alpha + n)^{-2} \left[\alpha^2(1-2\theta)^2 + n\theta(1-\theta) \right] \\ &= (2\alpha + n)^{-2} \left[(4\alpha^2 - n)\theta^2 - (4\alpha^2 - n)\theta + \alpha^2 \right] \end{aligned}$$

(set $\alpha^* = \sqrt{n}/2$)

$$= \frac{n/4}{(n + \sqrt{n})^2}$$

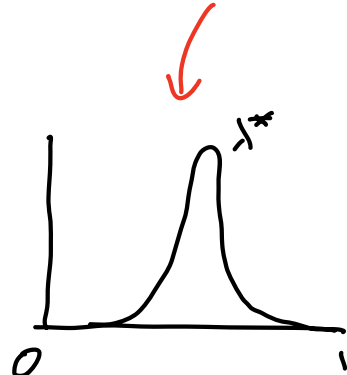
$$\Rightarrow \delta^*(X) = \frac{\sqrt{n}/2 + X}{\sqrt{n} + n}$$

$$r^* = n/4(n + \sqrt{n})^2$$

$$\Delta^* = \text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$$

$$\sigma_{\theta} \theta^{\sqrt{n}/2-1} (1-\theta)^{\sqrt{n}/2-1}$$

Q: why concentrated at $\theta = 1/2$?



Least Favorable Sequence

Sometimes there is no least favorable prior,

$X \sim N(\theta, 1)$: LF prior should spread mass everywhere, but that is not a proper prior.

Def: A sequence $\Lambda_1, \Lambda_2, \dots$ is LF if
$$r_{\Lambda_n} \rightarrow \sup_{\Lambda} r_{\Lambda}$$

Thm: Suppose $\Lambda_1, \Lambda_2, \dots$ is a prior sequence and δ satisfies
$$\sup_{\theta} R(\theta; \delta) = \lim_n r_{\Lambda_n}$$

Then a) δ is minimax

b) $\Lambda_1, \Lambda_2, \dots$ is LF

Proof a) Other est. $\tilde{\delta}$. Then $\forall n$,

$$\begin{aligned} \sup_{\theta} R(\theta; \tilde{\delta}) &\geq \int R(\theta; \tilde{\delta}) d\Lambda_n(\theta) \\ &\geq r_{\Lambda_n} \end{aligned}$$

$$\begin{aligned} \Rightarrow \sup_{\theta} R(\theta; \tilde{\delta}) &\geq \sup_n r_{\Lambda_n} \\ &\geq \lim_n r_{\Lambda_n} \\ &= \sup_{\theta} R(\theta; \delta) \end{aligned}$$

b) Prior Λ

$$r_\Lambda = \int R(\theta; \delta_\Lambda) d\Lambda(\theta)$$

$$\leq \int R(\theta; \delta) d\Lambda(\theta)$$

$$\leq \sup_{\theta} R(\theta; \delta)$$

$$= \lim_n r_{\Lambda_n}$$

□

Ex $X \sim N_d(0, I_d)$. Est. θ , use MSE

$$\delta_0(x) = x \quad \text{unbiased, } MSE = d$$

$$\xi = \frac{1}{1+n}$$

Try prior $\Lambda_n: \theta_i \stackrel{iid}{\sim} N(0, n) \Rightarrow \delta_{\Lambda_n}(x) = (1-\xi)X$

$$MSE(\theta; \delta_{\Lambda_n}) = \xi^2 \|\theta\|^2 + (1-\xi)^2 d$$

$$r_{\Lambda_n} = \mathbb{E} MSE(\theta)$$

$$= \xi^2 n d + (1-\xi)^2 d$$

$$= d \left(\frac{n}{(1+n)^2} + \frac{n^2}{(1+n)^2} \right)$$

$$\rightarrow d$$

So $\Lambda_1, \Lambda_2, \dots$ is LF

$\delta_0(x)$ is minimax (but inadmissible?)

$$r^* = d$$

Bounding minimax risk

Our theorem gives an idea of how to bound r^* for a problem:

Upper bound: If δ is any estimator then

$$r^* \leq \sup_{\theta} R(\theta; \delta) \quad (= \text{if } \delta \text{ minimax})$$

Lower bound: If Δ is any prior then

$$r^* \geq \int R(\theta; \delta_{\Delta}) d\Delta(\theta) \quad (= \text{if } \Delta \text{ LF})$$

Minimax estimators are very hard to find but minimax bounds are often used in stat theory to characterize hardness (esp. lower)

Ex: Propose practical estimator δ , find Δ for which r_{Δ} close to $\sup_{\theta} R(\theta; \delta)$ (or same rate, or cugs asymptotically)

$\Rightarrow$ Conclude δ can't be improved "much" (*)

Ex: Quantify hardness of a problem by its minimax rate in some asy. regime.

Caveat: A problem might be easy throughout most of par. space but very hard in some bizarre corner you never encounter in practice!